

Sérgio Alves

Full Stack Developer

Full Stack Development • APIs • Databases • Cloud • AI Integration

A selection of full-stack projects focused on building complete web products, from React interfaces and Node.js APIs to authentication, databases, external integrations and cloud deployment.

CORE STACK

React • TypeScript • JavaScript • Node.js • Express • PostgreSQL • MongoDB • REST APIs • Git • GitHub • Vercel

TECH INTEGRATIONS

AI Integration • Stripe • JWT • Neon • MongoDB Atlas



02 • CoupleAdventure

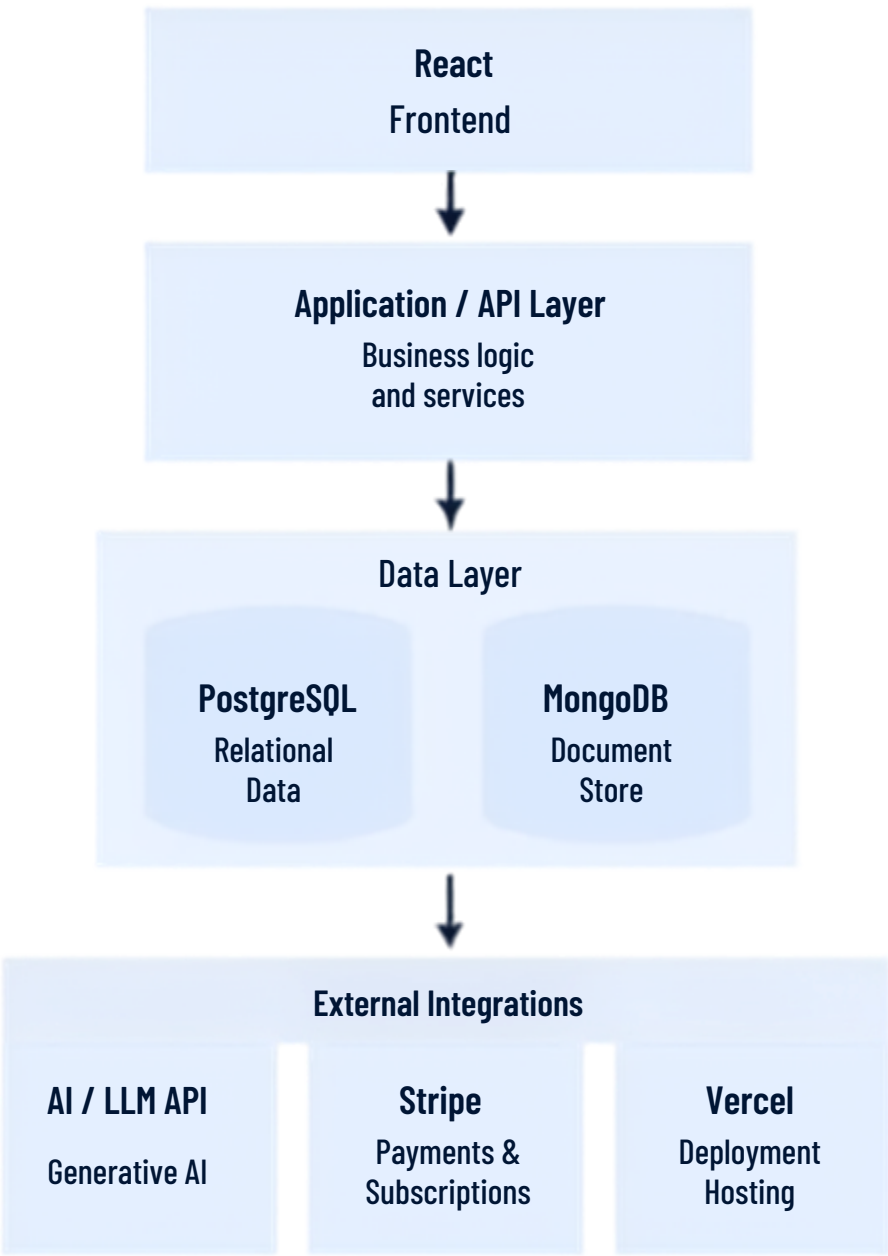
AI-Powered Interactive RPG Experience for Couples

A web platform where two people embark on an interactive RPG adventure. The story evolves in real time through AI, adapting to every choice you make together.

PRODUCT CONCEPT



TECHNICAL ARCHITECTURE



KEY FEATURES

- Account creation and management
- Adventures for two players
- AI-powered interactive RPG narrative
- Choices that impact the story
- Adventure state persistence and session history
- Subscriptions and payments via Stripe
- Full integration of frontend, backend, databases and external services
- Deployment on Vercel

ENGINEERING HIGHLIGHTS

- Real AI / LLM integration in a product
- Persistent state for continuous storytelling
- Hybrid architecture: PostgreSQL for relational data, MongoDB for documents
- Stripe for payments and subscriptions
- End-to-end product development, including authentication and deployment

TECH STACK

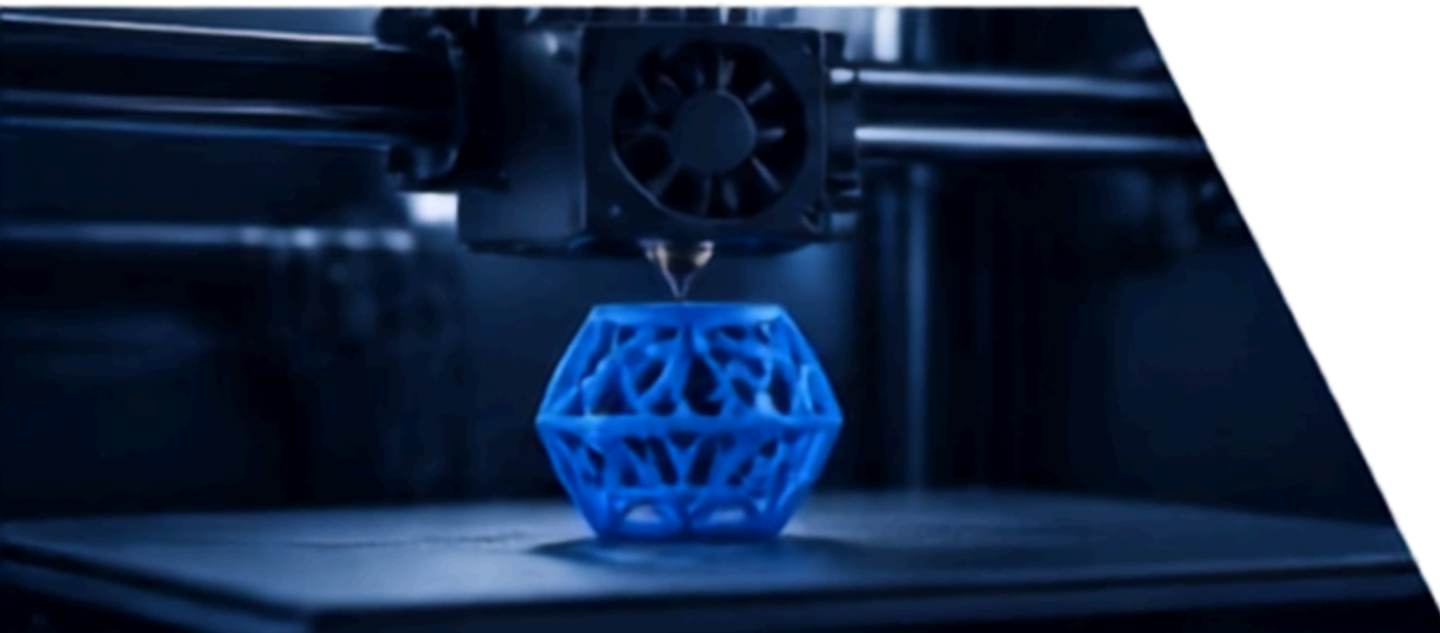
React | TypeScript | Node.js | PostgreSQL | MongoDB | AI/LLM API | Stripe | Vercel

Live Application
<https://coupleadventure.vercel.app>

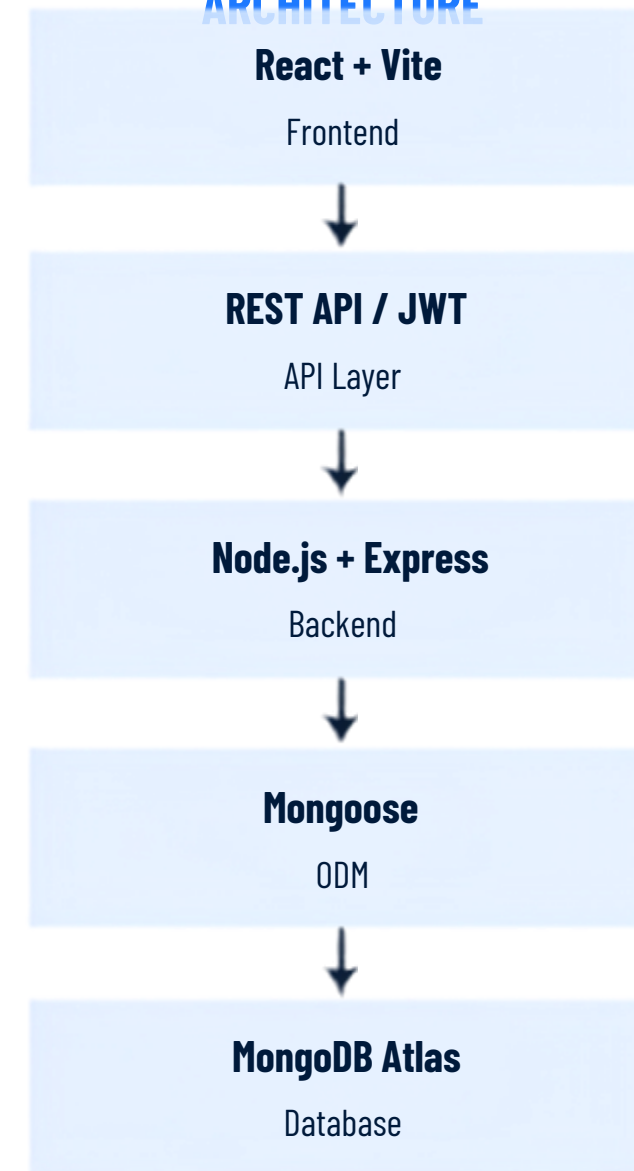
03 • 3D Printing Cost & Production Management Platform

A full-stack application for 3D printing cost calculation and production management.

A complete product with real business rules for managing parts, materials, stock, costs and pricing plans.



TECHNICAL ARCHITECTURE



KEY FEATURES

- JWT authentication with bcrypt
- User-Specific Calculation Settings
- Material, stock and part management, including creation, editing, duplication, archiving and deletion
- Cost Calculator & Dashboard
- Free/Maker/Pro plans with server-side permissions
- Administrative area
- CSV export
- Responsive interface with dark/light mode

ENGINEERING HIGHLIGHTS

- Clear separation between frontend, API and database
- Secrets protected from the browser
- Deployment on Vercel
- MongoDB Atlas for cloud persistence

STACK

React Vite TypeScript Node.js Express MongoDB Atlas Mongoose JWT bcrypt Vercel

Live Application

<https://calc3dprice.vercel.app>

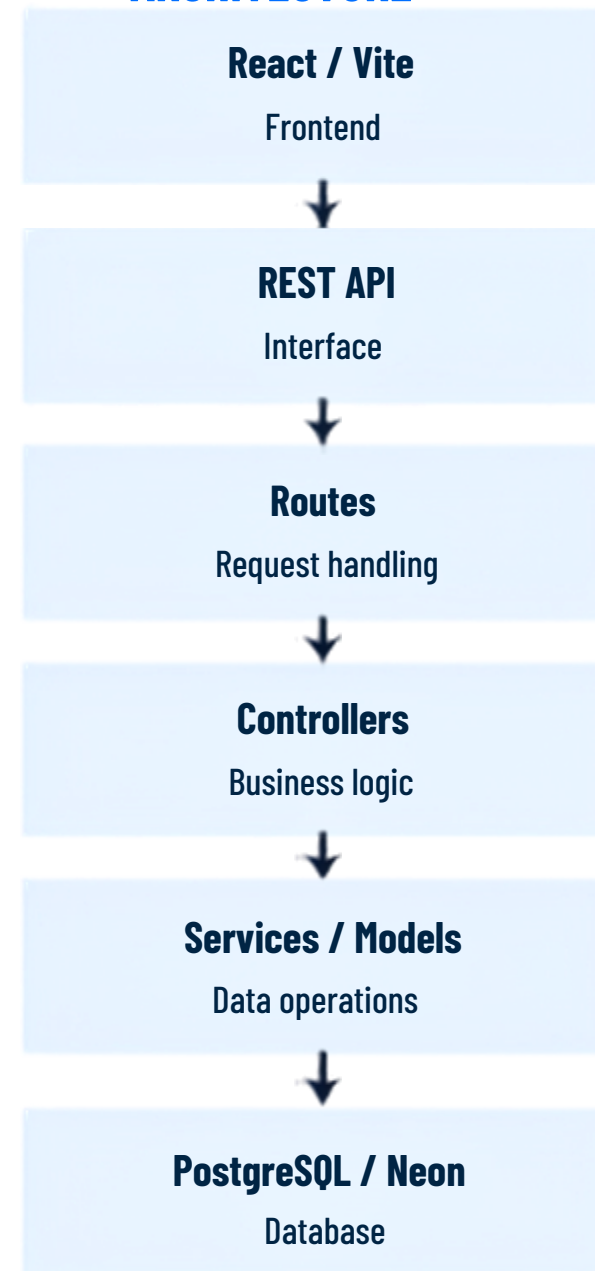
04 •

ClubScout – Sports Club Sales CRM

A structured CRM that turns a real sales process into an application with business rules, user roles and collaboration for club prospecting.



TECHNICAL ARCHITECTURE



KEY FEATURES

Club, contact, demo, attachment and shared-note management

Dashboard with statistics and persistent notifications for relevant changes

Password recovery via SMTP

JWT authentication with user roles, access levels and permissions

STACK

React	Vite	Node.js	Express
PostgreSQL		Neon	JWT
REST APIs		Vercel	

Source Code

<https://github.com/SerAlves16/CRM-clubscout>

05 •

White-Label Sports Club Management Platform

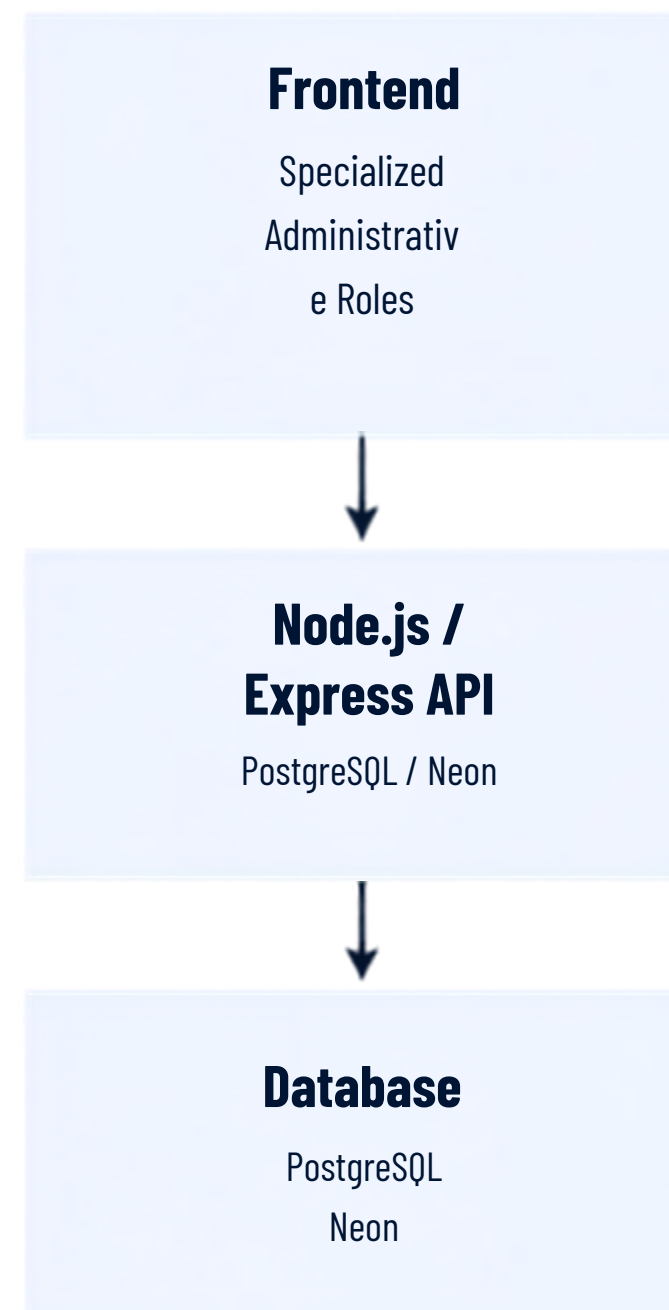
Adaptable platform for sports clubs, combining public presence with powerful administrative tools.

KEY FEATURES

Public website, news and sports sections

Membership management and fees; ticketing, store and member dashboard
Sponsors and member requests; social media planning; engagement analytics

Admin | Ticketing | Store | Communication | Finance | Member



ENGINEERING HIGHLIGHTS

JWT + bcrypt for security

Server-side authorization; PostgreSQL transactions for tickets and orders

Backend stock and pricing logic
Flexible white-label architecture

Live Application

<https://platclubabase.vercel.app>

TECH STACK

React 18

TypeScript

Vite

Tailwind

Node.js

Express

PostgreSQL

Neon

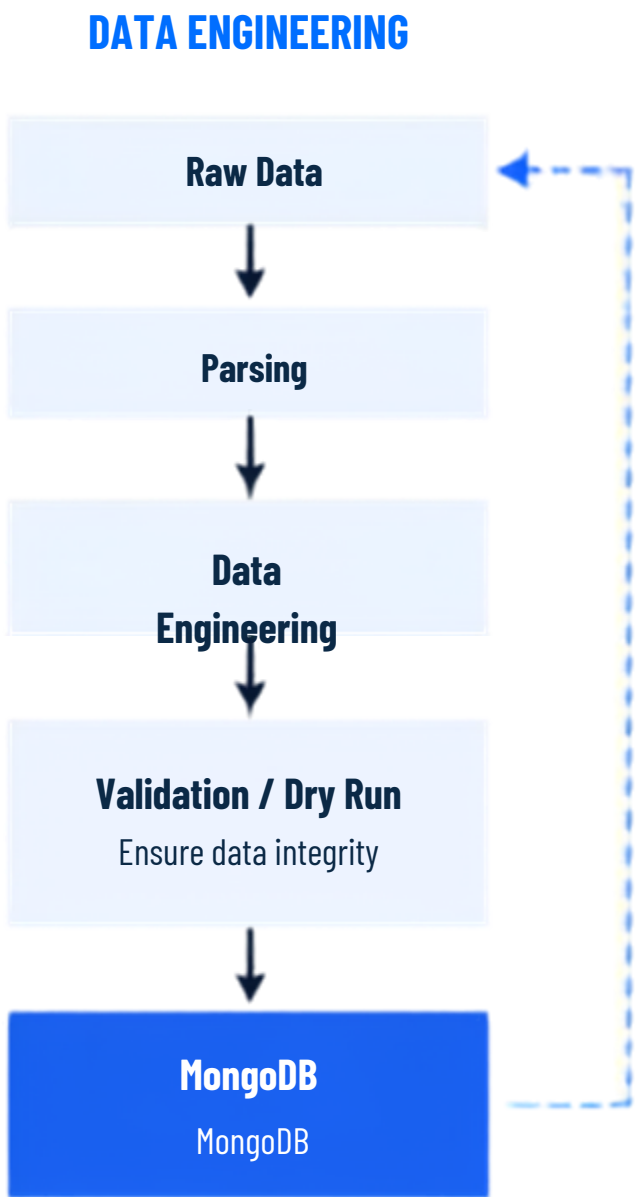
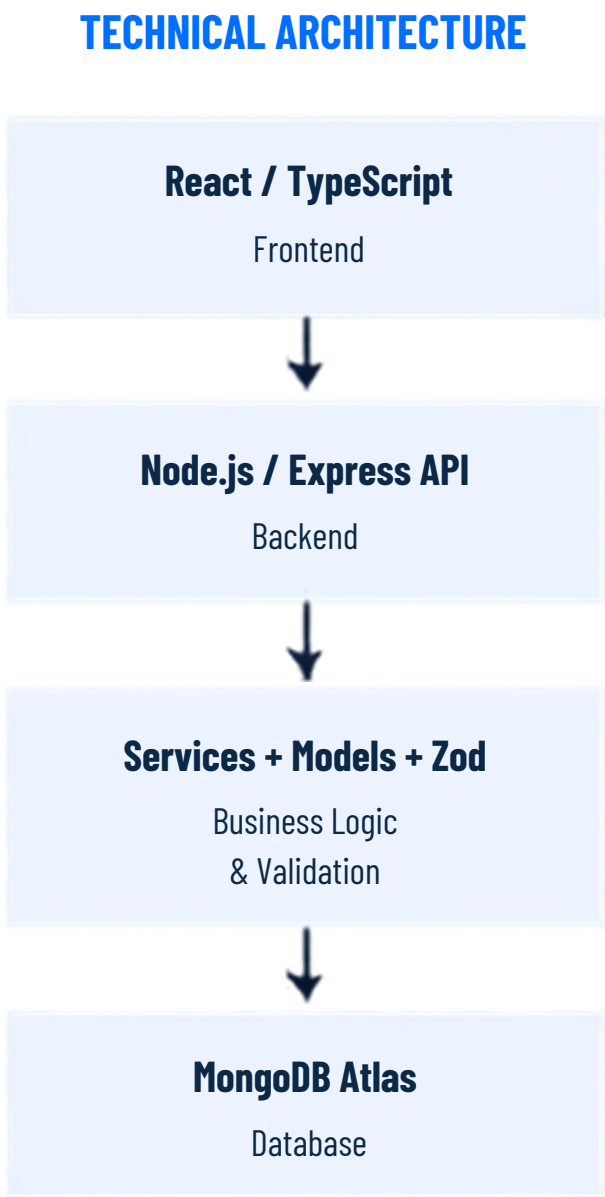
JWT

Vercel

06 • D&D 5e Character Builder

Full Stack Character Creation & Management Platform

A full-stack platform for creating and managing D&D 5e characters, with a layered API architecture, user data isolation and strict validation.



- KEY FEATURES
- Layered/MVC API architecture
 - JWT authentication
 - Rate limiting
 - Zod validation
 - User data isolation
 - CRUD operations and duplication
 - inventory, spells, items, classes, subclasses, backgrounds, feats, skills and languages

- DATA ENGINEERING HIGHLIGHTS
- Import and normalize large game-data catalogues while preserving raw data, stable identifiers and idempotent processes.
 - Dry-run validation before database changes.

07 • Beyond the Projects

Engineering Approach

A practical, end-to-end approach that turns problems into reliable, secure and scalable products.

01

Problem First

Understand the problem, users and business rules before choosing the implementation.

02

Full Stack Ownership

Build across frontend, APIs, authentication, business logic and persistence.

03

API & Data

Design REST APIs and work with relational and document databases including PostgreSQL and MongoDB.

04

Security by Design

Apply authentication, authorization, credential protection, hashing, validation and user data isolation.

05

Deploy & Iterate

Deploy and iterate using cloud/serverless infrastructure such as Vercel, Neon and MongoDB Atlas.

06

AI as a Product Component

Integrate AI/LLMs into applications as part of the product architecture and user experience.